



Exercise Sheet 09

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Due: July 14, 2025

Total points: 10

Please upload your solutions to WueCampus as a scanned document (image format or pdf), a typesetted PDF document, and/or as a jupyter notebook.

1. Detailed balance condition

- (a) Show that a Markov chain with a symmetric transition matrix fulfills the detailed balance condition. 2P

(Hint: Calculate the stationary probability first)

- (b) What can be said about the stationary distribution of this Markov Chain? 1P

2. Information Theory and Data Compression

- (a) Demonstrate the additivity of the information measure by showing that the information gained from observing the collective outcome of N independent events, each with probabilities p_i for $i = 1$ to N , is equal to the sum of the information gained from observing each event individually and independently, regardless of the order in which they are observed. 2P

(Hint: The information gain measure assigns $\log_2(p)$ bits to an event with a probability of p .)

- (b) Calculate the entropy, measured in bits, for each of the following random variables: 2P

- a) The pixel values in an image, where the possible grey values range from 0 to 255, and each value has an equal probability of occurrence.
- b) Humans classified based on whether they are mammals or not.
- c) Gender in a tri-sexed insect population, where the three genders have probabilities of occurrence equal to $\frac{1}{4}$, $\frac{1}{4}$, and $\frac{1}{2}$.
- d) A population of individuals classified as either older or not older than the median age of the population.

- (c) If we consider that women who surpass the age of 80 outnumber men in the same age range by a ratio of three to one, the question asks how much information, measured in bits, is obtained by discovering that a person who lives past 80 is male? 3P

(Hint: Consider that $p(\text{female}|\text{old}) + p(\text{male}|\text{old}) = 1$)